Math 4441: Probability and Statistics

Prof. Muhammad Mahbub Alam, PhD

Department of Computer Science and Engineering

Islamic University of Technology

Chapter 5
Chapter 6: Random Vectors and Gaussian Random Variables
Lectures 24

Math 4441: Probability and Statistics

References:

1. Goodman and Yates – Introduction to Probability and Stochastic Processes, 3rd Edition
2. Scott Miller and Donald Childers – Probability and Random Processes, 2nd Edition
3. Bertsekas and Tsitsiklis – Introduction to Probability, 2nd Edition

Vector Notations

Definition 5.2. *Random Vector.* A random vector is a column vector $\mathbf{X} = [X_1, X_2, \ldots, X_n]^T$, where each X_i is a random variable.

A random variable is a random vector with $n = 1$. The sample values of the components of a random vector constitute a column vector as well.

Definition 5.3. *Sapmle-Value Vector.* The sample values of the components of a random vector is a column vector, represented by the bold-face small letter $\mathbf{x} = [x_1, x_2, \ldots, x_n]^{\mathrm{T}}$, is defined as the sample-value vector. The i-th component of the vector $\mathbf{x}$, x_i, is a sample value of the random variable X_i. $\square$

Probability Models of Random Vectors

Definition 5.4. *CDF of a Random Vector.* The CDF of a random vector $\mathbf{X}$, for the sample-value vector $\mathbf{x}$, denoted as $F_{\mathbf{X}}(\mathbf{x})$, is given by

$$
\begin{aligned}
F_{\mathbf{X}}(\mathbf{x}) &= P[\mathbf{X} \leq \mathbf{x}] \\
&= F_{X_1, X_2, \ldots, X_n}(x_1, x_2, \ldots, x_n) \\
&= P[X_1 \leq x_1, X_2 \leq x_2, \ldots, X_n \leq x_n] \qquad (5.24)
\end{aligned}
$$

Definition 5.5. (*PMF of a Random Vector*:) The PMF of a discrete random vector $\mathbf{X}$, for the sample-value vector $\mathbf{x}$, denoted as $P_{\mathbf{X}}(\mathbf{x})$, is given by

$$
\begin{aligned}
P_{\mathbf{X}}(\mathbf{x}) &= P[\mathbf{X} = \mathbf{x}] \\
&= P_{X_1, X_2, \ldots, X_n}(x_1, x_2, \ldots, x_n) \\
&= P[X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n] \qquad (5.25)
\end{aligned}
$$

Definition 5.6. *PDF of a random vector*: The PDF of a continuous random vector $\mathbf{X}$, for the sample-value vector $\mathbf{x}$, is the multivariate joint PDF of the component random variables $X_1, X_2, \ldots, X_n$, and is given by

$$f_{\mathbf{X}}(\mathbf{x}) = f_{X_1,X_2,\ldots,X_n}(x_1, x_2, \ldots, x_n). \tag{5.26}$$

Joint Probability Models of Two Random Vectors

Definition 5.7. *Joint PMF of two Random Vectors.* The joint PMF of two discrete random vectors, X with n components and Y with m components, $P_{\mathbf{X},\mathbf{Y}}(\mathbf{x},\mathbf{y})$, is the multivariate joint PMF of $(n+m)$ discrete random variables and is given by

$$P_{\mathbf{X},\mathbf{Y}}(\mathbf{x},\mathbf{y}) = P_{X_1,\ldots,X_n,Y_1,\ldots,Y_m}(x_1,\ldots,x_n,y_1,\ldots,y_m). \tag{5.27}$$

Definition 5.8. *Joint PDF of two Random Vectors.* The joint PDF of two continuous random vectors, X with n components and Y with m components, $f_{\mathbf{X},\mathbf{Y}}(\mathbf{x},\mathbf{y})$, is the multivariate joint PMF of $(n+m)$ continuous random variables and is given by

$$f_{\mathbf{X},\mathbf{Y}}(\mathbf{x},\mathbf{y}) = f_{X_1,\ldots,X_n,Y_1,\ldots,Y_m}(x_1,\ldots,x_n,y_1,\ldots,y_m). \tag{5.28}$$

Definition 5.9. *Joint CDF of two Random Vectors.* The joint CDF of two discrete/continuous random vectors, X with n components and Y with m components, $F_{\mathbf{X},\mathbf{Y}}(\mathbf{x},\mathbf{y})$, is the multivariate joint CDF of $(n+m)$ continuous random variables and is given by

$$F_{\mathbf{X},\mathbf{Y}}(\mathbf{x},\mathbf{y}) = F_{X_1,\ldots,X_n,Y_1,\ldots,Y_m}(x_1,\ldots,x_n,y_1,\ldots,y_m). \tag{5.29}$$

Example 5.6. Random vector $\mathbf{X}$ has the PDF

$$f_{\mathbf{X}}(\mathbf{x}) = \begin{cases} 6e^{-\mathbf{a}^{\mathsf{T}}\mathbf{x}}, & \mathbf{x} > 0 \\ 0, & \text{otherwise.} \end{cases}$$

where $\mathbf{a}^{\mathsf{T}} = [1 \quad 2 \quad 3]$. What is the CDF of $\mathbf{X}$.

Solution. Because $\mathbf{a}$ has three components, we assume that $\mathbf{X}$ is also a three dimensional vector. Now,

$$\mathbf{a}^{\mathrm{T}}\mathbf{x} = x_1 + 2x_2 + 3x_3, \tag{5.30}$$

and the PDF can be given by

$$f_{\mathbf{X}}(\mathbf{x}) = 6e^{-x_1-2x_2-3x_3}, \quad x_1, x_2, x_3 > 0 \tag{5.31}$$

Finally, the CDF of the vector $\mathbf{X}$ can be found by integrating the PDF of the vector $\mathbf{X}$ over x_1, x_2 and x_3.

$$
\begin{aligned}
F_{\mathbf{X}}(\mathbf{x}) &= \int_{-\infty}^{x_3}\int_{-\infty}^{x_2}\int_{-\infty}^{x_1} 6e^{-x_1-2x_2-3x_3} dx_1 dx_2 dx_3 \\
&= \int_{0}^{x_3}\int_{0}^{x_2}\int_{0}^{x_1} 6e^{-x_1}e^{-2x_2}e^{-3x_3} dx_1 dx_2 dx_3 \\
&= \int_{0}^{x_3}\int_{0}^{x_2} 6e^{-2x_2}e^{-3x_3}\left[-e^{-x_1}\right]_0^{x_1} dx_2 dx_3 \\
&= 6(1-e^{-x_1})\int_{0}^{x_3} e^{-3x_3}\left[-\frac{1}{2}e^{-2x_2}\right]_0^{x_2} dx_3 \\
&= (1-e^{-x_1})(1-e^{-2x_2})(1-e^{-3x_3})
\end{aligned}
$$

Independent Random Vectors

Two random vectors $\mathbf{X}$ and $\mathbf{Y}$ are independent, if for two discrete random vectors, the following is true

$$P_{\mathbf{X},\mathbf{Y}}(\mathbf{x},\mathbf{y}) = P_{\mathbf{X}}(\mathbf{x})P_{\mathbf{Y}}(\mathbf{y}),$$

and for two continuous random vectors, the following relation is true

$$f_{\mathbf{X},\mathbf{Y}}(\mathbf{x},\mathbf{y}) = f_{\mathbf{X}}(\mathbf{x})f_{\mathbf{Y}}(\mathbf{y}).$$

Example 5.7. Random variables $Y_1, Y_2, \ldots, Y_4$ have the following multivariate joint PDF:

$$f_{Y_1,Y_2,\ldots,Y_4}(y_1, y_2, \ldots y_n) = \begin{cases} 4, & \text{for } 0 \le y_1 \le y_2 \le 1, 0 \le y_3 \le y_4 \le 1 \\ 0, & \text{otherwise.} \end{cases}$$

Let $\mathbf{V} = \begin{bmatrix} Y_1 & Y_4 \end{bmatrix}'$ and $\mathbf{W} = \begin{bmatrix} Y_2 & Y_3 \end{bmatrix}'$. Are $\mathbf{V}$ and $\mathbf{W}$ independent random vectors?

$\mathbf{V}$ are $V_1 = Y_1$, and $V_2 = Y_4$.

$$P_{\mathbf{V},\mathbf{W}}(v, w) = \begin{cases} 4, & \text{for } 0 \le v_1 \le w_1 \le 1, 0 \le w_2 \le v_2 \le 1 \\ 0, & \text{otherwise.} \end{cases}$$

$$f_{\mathbf{V}}(\mathbf{v}) = \begin{cases} 4(1-v_1)v_2 & 0 \le v_1, v_2 \le 1, \\ 0 & \text{otherwise,} \end{cases}$$

$$f_{\mathbf{W}}(\mathbf{w}) = \begin{cases} 4w_1(1-w_2) & 0 \le w_1, w_2 \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

$$f_{\mathbf{V}}(\mathbf{v})\, f_{\mathbf{W}}(\mathbf{w}) = \begin{cases} 16(1-v_1)v_2 w_1(1-w_2) & 0 \le v_1, v_2, w_1, w_2 \le 1, \\ 0 & \text{otherwise,} \end{cases}$$

V and W are not independent

Expected Value Vector and Correlation Matrix

Definition 5.10. *Expected value vector*: The expected value vector is the expected value of a random vector and its i-th component is the expected value of the i-th component random variable of the random vector.

If $\mathbf{X}$ is a random vector with n component random variables, $X_1, X_2, \ldots, X_n$, then the expected value vector of $\mathbf{X}$, $E[\mathbf{X}]$, is given by

$$E[\mathbf{X}] = \mu_{\mathbf{X}} = \begin{bmatrix} E[X_1] & E[X_2] & \cdots & E[X_n] \end{bmatrix}^T \tag{5.36}$$

For example, the expected value vector of the random vector, $\mathbf{X} = [X_1 \quad X_2 \quad X_3]^T$ is given by

$$E[\mathbf{X}] = \begin{bmatrix} E[X_1] & E[X_2] & E[X_3] \end{bmatrix}^T. \tag{5.37}$$

Covariance of Random Vector

$$\mathrm{Cov}[\mathbf{X}] = E[(\mathbf{X} - \mu_{\mathbf{X}})(\mathbf{X} - \mu_{\mathbf{X}})^T].$$

$$\begin{aligned} Y_{i,j} &= E[(X_i - E[X_i])(X_j - E[X_j])] \\ &= \mathrm{Cov}[X_i, X_j]. \end{aligned}$$

Definition 5.11. *Covariance Matrix.* The covariance matrix of a random vector $\mathbf{X}$, $Cov[\mathbf{X}]$, represents covariance of every pair of components random variables of the random vector, adn is given by

$$Cov[\mathbf{X}] = \begin{bmatrix} \mathrm{Var}[X] & Cov[X_1, X_2] & \cdots & Cov[X_1, X_n] \\ Cov[X_2, X_1] & Var[X_2, X_2] & \cdots & Cov[X_2, X_n] \\ \vdots & \vdots & \ddots & \vdots \\ Cov[X_n, X_1] & Cov[X_n, X_2] & \cdots & Var[X_n, X_n] \end{bmatrix}$$

$$
\begin{aligned}
\mathrm{Cov}[\mathbf{X}] &= E[(\mathbf{X}-\mu_{\mathbf{X}})(\mathbf{X}-\mu_{\mathbf{X}})^T] \\
&= E[\mathbf{X}\mathbf{X}^T - \mathbf{X}\mu_{\mathbf{X}}^T - \mu_{\mathbf{X}}\mathbf{X}^T + \mu_{\mathbf{X}}\mu_{\mathbf{X}}^T] \\
&= E[\mathbf{X}\mathbf{X}^T] - E[\mathbf{X}\mu_{\mathbf{X}}^T] - E[\mu_{\mathbf{X}}\mathbf{X}^T] + E[\mu_{\mathbf{X}}\mu_{\mathbf{X}}^T] \\
&= E[\mathbf{X}\mathbf{X}^T] - E[\mathbf{X}]\mu_{\mathbf{X}}^T - \mu_{\mathbf{X}}E[\mathbf{X}^T] + \mu_{\mathbf{X}}\mu_{\mathbf{X}}^T \\
&= E[\mathbf{X}\mathbf{X}^T] - \mu_{\mathbf{X}}\mu_{\mathbf{X}}^T - \mu_{\mathbf{X}}\mu_{\mathbf{X}}^T] + \mu_{\mathbf{X}}\mu_{\mathbf{X}}^T \\
&= E[\mathbf{X}\mathbf{X}^T] - \mu_{\mathbf{X}}\mu_{\mathbf{X}}^T
\end{aligned}
$$

Correlation Matrix

Definition 5.12. *Correlation Matrix*: The correlation of a random vector $\mathbf{X}$ with n components is an $n \times n$ matrix, $R_{\mathbf{X}}$, whose (i, j)-th elements $R_{\mathbf{X}}(i, j) = E[X_i, X_j]$, and is given by

$$R_{\mathbf{X}} = \begin{bmatrix} E[X_1^2] & E[X_1, X_2] & \cdots & E[X_1, X_n] \\ E[X_2, X_1] & E[X_2^2] & \cdots & E[X_2, X_n] \\ \vdots & \vdots & \ddots & \vdots \\ E[X_n, X_1] & E[X_n, X_2] & \cdots & E[X_n^2] \end{bmatrix}$$

Finally, the covariance of a random vector $\mathbf{X}$ in terms of its correlation $R_{\mathbf{X}}$ and expectation $\mu_{\mathbf{X}}$ is given by

$$Cov[\mathbf{X}] = R_{\mathbf{X}} - \mu_{\mathbf{X}} \mu_{\mathbf{X}}^T.$$

Example 5.8. The two dimensional random vector $\mathbf{X} = [X_1 \quad X_2]^T$ has the following PDF:

$$f_{\mathbf{X}}(\mathbf{x}) = \begin{cases} 2, & \text{for } 0 \leq x_1 \leq x_2 \leq 1 \\ 0, & \text{otherwise.} \end{cases}$$

Find the expected value $E[\mathbf{X}]$, correlation matrix $R_{\mathbf{X}}$, and coariance matrix $\mathrm{Cov}_{\mathbf{X}}$ for the random vector $\mathbf{X}$.

$$
\begin{aligned}
f_{X_1}(x_1) &= \int_{x_l}^{1} 2dx_2 \\
&= 2(1 - x_1), \qquad 0 \leq x_1 \leq 1.
\end{aligned}
$$

$$
\begin{aligned}
f_{X_2}(x_2) &= \int_{0}^{x_2} 2dx_1 \\
&= 2x_2, \qquad 0 \leq x_2 \leq 1.
\end{aligned}
$$

The expected values of the component random variables are given by

$$
\begin{aligned}
E[X_1] &= \int_0^1 x_1.2(1-x_1)dx_1 \\
&= 2\left[\frac{x_1^2}{2}\Big|_0^1 - \frac{x_1^3}{3}\Big|_0^1\right] \\
&= 2\left(\frac{1}{2}-\frac{1}{3}\right) \\
&= 2\times\frac{3-2}{6} \\
&= \frac{1}{3}.
\end{aligned}
$$

$$
\begin{aligned}
E[X_2] &= \int_0^1 x_2.2x_2 dx_2 \\
&= 2\frac{x_2^3}{3}\Big|_0^1 \\
&= \frac{2}{3}.
\end{aligned}
$$

Therefore, the expected value vector for $\mathbf{X}$ is

$$E[\mathbf{X}] = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} \end{bmatrix}^T$$

Now the second moments of the component random variables are given by

$$
\begin{aligned}
E[X_1^2] &= \int_0^1 x_1^2 . 2(1 - x_1) dx_1 \\
&= 2\left[\frac{x_1^3}{3}\Big|_0^1 - \frac{x_1^4}{4}\Big|_0^1 \right] \\
&= 2\left(\frac{1}{3} - \frac{1}{4} \right) \\
&= 2 \times \frac{4-3}{12} \\
&= \frac{1}{6}.
\end{aligned}
$$

$$
\begin{aligned}
E[X_2^2] &= \int_0^1 x_2^2.2x_2 dx_2 \\
&= 2\frac{x_2^4}{4}\Big|_0^1 \\
&= \frac{1}{2}.
\end{aligned}
$$

And the expected value of the product of the two component random variables is given by

$$
\begin{aligned}
E[X_1X_2] &= \int_0^1 \int_0^{x_2} x_1x_2 2dx_1dx_2 \\
&= \int_0^1 x_2.\frac{x_1^2}{2}\Big|_0^{x_2} dx_2 \\
&= \int_0^1 x_2.x_2^2 dx_2 \\
&= \frac{x_2^4}{4}\Big|_0^1 \\
&= \frac{1}{4}.
\end{aligned}
$$

Therefore, the correlation matrix fo the random vector $\mathbf{X}$ is

$$R_{\mathbf{X}} = \begin{bmatrix} \frac{1}{6} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{2} \end{bmatrix}.$$

Finally, the covariance matrix of the random vector $\mathbf{X}$ is given by

$$\begin{aligned} \mathrm{Cov}_{\mathbf{X}} &= R_{\mathbf{X}} - \mu_{\mathbf{X}}\mu_{\mathbf{X}}^T \\ &= \begin{bmatrix} \frac{1}{6} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{2} \end{bmatrix} - \begin{bmatrix} \frac{1}{3} \\ \frac{2}{3} \end{bmatrix} - \begin{bmatrix} \frac{1}{3} & \frac{2}{3} \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{6} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{2} \end{bmatrix} - \begin{bmatrix} \frac{1}{9} & \frac{2}{9} \\ \frac{2}{9} & \frac{4}{9} \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{18} & \frac{1}{36} \\ \frac{1}{36} & \frac{1}{18} \end{bmatrix}. \end{aligned}$$

Cross-correlation and Cross-covariance of Random Vectors

The cross-correlation of a pair of random vectors $\mathbf{X}$ with n component random variables and $\mathbf{Y}$ with m component random variables is an $n \times m$ matrix $R_{\mathbf{X},\mathbf{Y}}$ with the (i, j)-th element $R_{\mathbf{X},\mathbf{Y}}(i, j) = E[X_i Y_j]$ which, in the vector form, is given by

$$\mathbf{R}_{\mathbf{XY}} = E[\mathbf{X}\mathbf{Y}^T].$$

Definition 5.13. *Cross-covariance*: The cross-covariance of a pair of random vectors $\mathbf{X}$ with n component random variables and $\mathbf{Y}$ with m component random variables is an $n \times m$ matrix $\mathbf{Cov}[\mathbf{X}, \mathbf{Y}]$ with the (i, j)-th element $\mathrm{Cov}[\mathbf{X}, \mathbf{Y}](i, j) = \mathrm{Cov}[X_i Y_j]$ which, in the vector form, is given by

$$\mathbf{Cov}[\mathbf{X}, \mathbf{Y}] = E[(\mathbf{X} - \boldsymbol{\mu}_{\mathbf{X}})(\mathbf{Y} - \boldsymbol{\mu}_{\mathbf{Y}})^T].$$

$\mathbf{X}$ is an n-dimensional random vector with expected value $\boldsymbol{\mu}_{\mathbf{X}}$, correlation $\mathbf{R}_{\mathbf{X}}$, and covariance $\mathbf{C}_{\mathbf{X}}$. The m-dimensional random vector $\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b}$, where $\mathbf{A}$ is an $m \times n$ matrix and $\mathbf{b}$ is an m-dimensional vector, has expected value $\boldsymbol{\mu}_{\mathbf{Y}}$, correlation matrix $\mathbf{R}_{\mathbf{Y}}$, and covariance matrix $\mathbf{C}_{\mathbf{Y}}$ given by

$$\boldsymbol{\mu}_{\mathbf{Y}} = \mathbf{A}\boldsymbol{\mu}_{\mathbf{X}} + \mathbf{b},$$
$$\mathbf{R}_{\mathbf{Y}} = \mathbf{A}\mathbf{R}_{\mathbf{X}}\mathbf{A}' + (\mathbf{A}\boldsymbol{\mu}_{\mathbf{X}})\mathbf{b}' + \mathbf{b}(\mathbf{A}\boldsymbol{\mu}_{\mathbf{X}})' + \mathbf{b}\mathbf{b}',$$
$$\mathbf{C}_{\mathbf{Y}} = \mathbf{A}\mathbf{C}_{\mathbf{X}}\mathbf{A}'.$$

Example 5.9. Given the expected value $\boldsymbol{\mu}_{\mathbf{X}}$, the correlation $\mathbf{R}_{\mathbf{X}}$, and the covariance $\mathbf{Cov}[\mathbf{X}]$ of random vector $\mathbf{X}$ in Example -.-, and $\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b}$, where

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 6 & 3 \\ 3 & 6 \end{bmatrix}$$

and

$$\mathbf{b} = \begin{bmatrix} 0 \\ -2 \\ -2 \end{bmatrix}.$$

Find the expected value $\boldsymbol{\mu}_{\mathbf{Y}}$, the correlation $\mathbf{R}_{\mathbf{Y}}$, adn the covariance $\mathbf{Cov}[\mathbf{Y}]$.

$$
\begin{aligned}
\boldsymbol{\mu}_{\mathbf{Y}} &= \mathbf{A}\boldsymbol{\mu}_{\mathbf{X}} + \mathbf{b} \\
&= \begin{bmatrix} 1 & 0 \\ 6 & 3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} \frac{1}{3} \\ \frac{2}{3} \end{bmatrix} + \begin{bmatrix} 0 \\ -2 \\ -2 \end{bmatrix} \\
&= \begin{bmatrix} \frac{1}{3} \\ 4 \\ 5 \end{bmatrix} + \begin{bmatrix} 0 \\ -2 \\ -2 \end{bmatrix} \\
&= \begin{bmatrix} \frac{1}{3} \\ 2 \\ 3 \end{bmatrix}.
\end{aligned}
$$

The correlation $\mathbf{R_X}$ is given by

$$\mathbf{R_Y} = \mathbf{A}\mathbf{R_X}\mathbf{A}^T + (\mathbf{A}\boldsymbol{\mu}_\mathbf{X})\mathbf{b}^T + \mathbf{b}(\mathbf{A}\boldsymbol{\mu}_\mathbf{X})^T + \mathbf{b}\mathbf{b}^T$$

$$= \begin{bmatrix} \frac{1}{6} & \frac{13}{12} & \frac{4}{3} \\ \frac{13}{12} & \frac{15}{2} & \frac{37}{4} \\ \frac{4}{3} & \frac{37}{4} & \frac{25}{2} \end{bmatrix}$$

$$
\begin{aligned}
\mathbf{Cov}[\mathbf{Y}] &= \mathbf{A}\mathbf{Cov}[\mathbf{X}]\mathbf{A}^T \\
&= \begin{bmatrix} 1 & 0 \\ 6 & 3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} \frac{1}{18} & \frac{1}{36} \\ \frac{1}{36} & \frac{1}{18} \end{bmatrix} \begin{bmatrix} 1 & 6 & 3 \\ 0 & 3 & 6 \end{bmatrix} \\
&= \begin{bmatrix} \frac{1}{18} & \frac{1}{36} \\ \frac{5}{12} & \frac{1}{3} \\ \frac{1}{3} & \frac{5}{12} \end{bmatrix} \begin{bmatrix} 1 & 6 & 3 \\ 0 & 3 & 6 \end{bmatrix} \\
&= \begin{bmatrix} \frac{1}{18} & \frac{5}{12} & \frac{1}{3} \\ \frac{5}{12} & \frac{7}{2} & \frac{13}{4} \\ \frac{1}{3} & \frac{13}{4} & \frac{7}{2} \end{bmatrix}.
\end{aligned}
$$

If $\mathbf{X}$ *is a continuous random vector and* $\mathbf{A}$ *is an invertible matrix, then* $\mathbf{Y} = \mathbf{AX}+\mathbf{b}$ *has PDF*

$$f_{\mathbf{Y}}(\mathbf{y}) = \frac{1}{|det(\mathbf{A})|} f_{\mathbf{X}}\left(\mathbf{A}^{-1}(\mathbf{y} - \mathbf{b})\right)$$

The vectors $\mathbf{X}$ *and* $\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b}$ *have cross-correlation* $\mathbf{R}_{\mathbf{XY}}$ *and cross-covariance* $\mathbf{C}_{\mathbf{XY}}$ *given by*

$$\mathbf{R}_{\mathbf{XY}} = \mathbf{R}_{\mathbf{X}}\mathbf{A}' + \mu_{\mathbf{X}}\mathbf{b}', \qquad \mathbf{C}_{\mathbf{XY}} = \mathbf{C}_{\mathbf{X}}\mathbf{A}'.$$

Example 8.9

Continuing Example 8.8 for random vectors $\mathbf{X}$ and $\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b}$, calculate

(a) The cross-correlation matrix $\mathbf{R}_{\mathbf{XY}}$ and the cross-covariance matrix $\mathbf{C}_{\mathbf{XY}}$.

(b) The correlation coefficients ρ_{Y_1,Y_3} and ρ_{X_2,Y_1}.

(a) Direct matrix calculation using Theorem 8.9 yields

$$\mathbf{R_{XY}} = \begin{bmatrix} 1/6 & 13/12 & 4/3 \\ 1/4 & 5/3 & 29/12 \end{bmatrix}; \qquad \mathbf{C_{XY}} = \begin{bmatrix} 1/18 & 5/12 & 1/3 \\ 1/36 & 1/3 & 5/12 \end{bmatrix}.$$

(a) Direct matrix calculation using Theorem 8.9 yields

$$\mathbf{R}_{\mathbf{XY}} = \begin{bmatrix} 1/6 & 13/12 & 4/3 \\ 1/4 & 5/3 & 29/12 \end{bmatrix}; \qquad \mathbf{C}_{\mathbf{XY}} = \begin{bmatrix} 1/18 & 5/12 & 1/3 \\ 1/36 & 1/3 & 5/12 \end{bmatrix}.$$

(b) Referring to Definition 5.6 and recognizing that $\mathrm{Var}[Y_i] = C_{\mathbf{Y}}(i,i)$, we have

$$\rho_{Y_1,Y_3} = \frac{\mathrm{Cov}[Y_1, Y_3]}{\sqrt{\mathrm{Var}[Y_1]\,\mathrm{Var}[Y_3]}} = \frac{C_Y(1,3)}{\sqrt{C_Y(1,1)C_Y(3,3)}} = 0.756 \tag{8.38}$$

Similarly,

$$\rho_{X_2,Y_1} = \frac{\mathrm{Cov}[X_2, Y_1]}{\sqrt{\mathrm{Var}[X_2]\,\mathrm{Var}[Y_1]}} = \frac{C_{\mathbf{XY}}(2,1)}{\sqrt{C_{\mathbf{X}}(2,2)C_{\mathbf{Y}}(1,1)}} = 1/2. \tag{8.39}$$

Definition 8.12 — Gaussian Random Vector

$\mathbf{X}$ *is the Gaussian* $(\boldsymbol{\mu}_\mathbf{X}, \mathbf{C}_\mathbf{X})$ *random vector with expected value* $\boldsymbol{\mu}_\mathbf{X}$ *and covariance* $\mathbf{C}_\mathbf{X}$ *if and only if*

$$f_\mathbf{X}(\mathbf{x}) = \frac{1}{(2\pi)^{n/2}[det(\mathbf{C}_\mathbf{X})]^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_\mathbf{X})' \mathbf{C}_\mathbf{X}^{-1} (\mathbf{x} - \boldsymbol{\mu}_\mathbf{X})\right)$$

A Gaussian random vector $\mathbf{X}$ *has independent components if and only if* $\mathbf{C}_{\mathbf{X}}$ *is a diagonal matrix.*

Example 8.10

Consider the outdoor temperature at a certain weather station. On May 5, the temperature measurements in units of degrees Fahrenheit taken at 6 AM, 12 noon, and 6 PM are all Gaussian random variables, X_1, X_2, X_3, with variance 16 degrees^2. The expected values are 50 degrees, 62 degrees, and 58 degrees respectively. The covariance matrix of the three measurements is

$$\mathbf{C}_\mathbf{X} = \begin{bmatrix} 16.0 & 12.8 & 11.2 \\ 12.8 & 16.0 & 12.8 \\ 11.2 & 12.8 & 16.0 \end{bmatrix}. \tag{8.45}$$

(a) Write the joint PDF of X_1, X_2 using the algebraic notation of Definition 5.10.
(b) Write the joint PDF of X_1, X_2 using vector notation.
(c) Write the joint PDF of $\mathbf{X} = \begin{bmatrix} X_1 & X_2 & X_3 \end{bmatrix}'$ using vector notation.

(a) First we note that X_1 and X_2 have expected values $\mu_1 = 50$ and $\mu_2 = 62$, variances $\sigma_1^2 = \sigma_2^2 = 16$, and covariance $\mathrm{Cov}[X_1, X_2] = 12.8$. It follows from Definition 5.6 that the correlation coefficient is

$$\rho_{X_1,X_2} = \frac{\mathrm{Cov}\left[X_1, X_2\right]}{\sigma_1 \sigma_2} = \frac{12.8}{16} = 0.8. \tag{8.46}$$

From Definition 5.10, the joint PDF is

$$f_{X_1,X_2}(x_1, x_2) = \frac{\exp\left[-\frac{(x_1 - 50)^2 - 1.6(x_1 - 50)(x_2 - 62) + (x_2 - 62)^2}{19.2}\right]}{60.3}.$$

(b) Let $\mathbf{W} = \begin{bmatrix} X_1 & X_2 \end{bmatrix}'$ denote a vector representation for random variables X_1 and X_2. From the covariance matrix $\mathbf{C_X}$, we observe that the 2×2 submatrix in the upper left corner is the covariance matrix of the random vector $\mathbf{W}$. Thus

$$\boldsymbol{\mu}_{\mathbf{W}} = \begin{bmatrix} 50 \\ 62 \end{bmatrix}, \qquad \mathbf{C_W} = \begin{bmatrix} 16.0 & 12.8 \\ 12.8 & 16.0 \end{bmatrix}. \tag{8.47}$$

We observe that $\det(\mathbf{C_W}) = 92.16$ and $\det(\mathbf{C_W})^{1/2} = 9.6$. From Definition 8.12, the joint PDF of $\mathbf{W}$ is

$$f_{\mathbf{W}}(\mathbf{w}) = \frac{1}{60.3} \exp\left(-\frac{1}{2}(\mathbf{w} - \boldsymbol{\mu}_{\mathbf{W}})^T \mathbf{C}_{\mathbf{W}}^{-1}(\mathbf{w} - \boldsymbol{\mu}_{\mathbf{W}})\right). \tag{8.48}$$

(c) Since $\boldsymbol{\mu}_{\mathbf{X}} = \begin{bmatrix} 50 & 62 & 58 \end{bmatrix}'$ and $\det(\mathbf{C}_{\mathbf{X}})^{1/2} = 22.717$, $\mathbf{X}$ has PDF

$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{357.8} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_{\mathbf{X}})^T \mathbf{C}_{\mathbf{X}}^{-1} (\mathbf{x} - \boldsymbol{\mu}_{\mathbf{X}})\right). \tag{8.49}$$

Given an n-dimensional Gaussian random vector $\mathbf{X}$ *with expected value* $\boldsymbol{\mu}_{\mathbf{X}}$ *and covariance* $\mathbf{C}_{\mathbf{X}}$, *and an* $m \times n$ *matrix* $\mathbf{A}$ *with rank*$(\mathbf{A}) = m$,

$$\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b}$$

is an m-dimensional Gaussian random vector with expected value $\boldsymbol{\mu}_{\mathbf{Y}} = \mathbf{A}\boldsymbol{\mu}_{\mathbf{X}} + \mathbf{b}$ *and covariance* $\mathbf{C}_{\mathbf{Y}} = \mathbf{A}\mathbf{C}_{\mathbf{X}}\mathbf{A}'$.

Example 8.11

Continuing Example 8.10, use the formula $Y_i = (5/9)(X_i - 32)$ to convert the three temperature measurements to degrees Celsius.

(a) What is $\boldsymbol{\mu}_{\mathbf{Y}}$, the expected value of random vector $\mathbf{Y}$?

(b) What is $\mathbf{C}_{\mathbf{Y}}$, the covariance of random vector $\mathbf{Y}$?

(c) Write the joint PDF of $\mathbf{Y} = \begin{bmatrix} Y_1 & Y_2 & Y_3 \end{bmatrix}'$ using vector notation.

(a) In terms of matrices, we observe that $\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{b}$ where

$$\mathbf{A} = \begin{bmatrix} 5/9 & 0 & 0 \\ 0 & 5/9 & 0 \\ 0 & 0 & 5/9 \end{bmatrix}, \qquad \mathbf{b} = -\frac{160}{9}\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}. \tag{8.55}$$

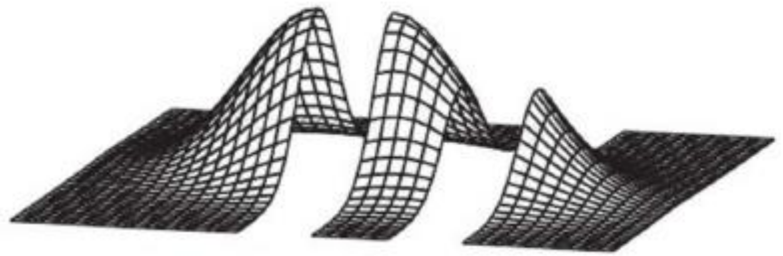
(b) Since $\boldsymbol{\mu}_{\mathbf{X}} = \begin{bmatrix} 50 & 62 & 58 \end{bmatrix}'$, from Theorem 8.11,

$$\boldsymbol{\mu}_{\mathbf{Y}} = \mathbf{A}\boldsymbol{\mu}_{\mathbf{X}} + \mathbf{b} = \begin{bmatrix} 10 \\ 50/3 \\ 130/9 \end{bmatrix}. \tag{8.56}$$

(c) The covariance of $\mathbf{Y}$ is $\mathbf{C}_{\mathbf{Y}} = \mathbf{A}\mathbf{C}_{\mathbf{X}}\mathbf{A}'$. We note that $\mathbf{A} = \mathbf{A}' = (5/9)\mathbf{I}$ where $\mathbf{I}$ is the 3×3 identity matrix. Thus $\mathbf{C}_{\mathbf{Y}} = (5/9)^2\mathbf{C}_{\mathbf{X}}$ and $\mathbf{C}_{\mathbf{Y}}^{-1} = (9/5)^2\mathbf{C}_{\mathbf{X}}^{-1}$. The PDF of $\mathbf{Y}$ is

$$f_{\mathbf{Y}}(\mathbf{y}) = \frac{1}{24.47}\exp\left(-\frac{81}{50}(\mathbf{y} - \boldsymbol{\mu}_{\mathbf{Y}})^T\mathbf{C}_{\mathbf{X}}^{-1}(\mathbf{y} - \boldsymbol{\mu}_{\mathbf{Y}})\right). \tag{8.57}$$

If $f_{XY}(x, y)$ is a jointly Gaussian PDF for random variables X and Y, then



i. Both X and Y have the normal or Gaussian PDF, i.e., the marginal PDF of X and Y are:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma_X^2}} e^{-\frac{(x-\mu_X)^2}{2\sigma_X^2}}, \quad -\infty < x < +\infty$$

$$f_Y(y) = \frac{1}{\sqrt{2\pi\sigma_Y^2}} e^{-\frac{(y-\mu_Y)^2}{2\sigma_Y^2}}, \quad -\infty < y < +\infty.$$

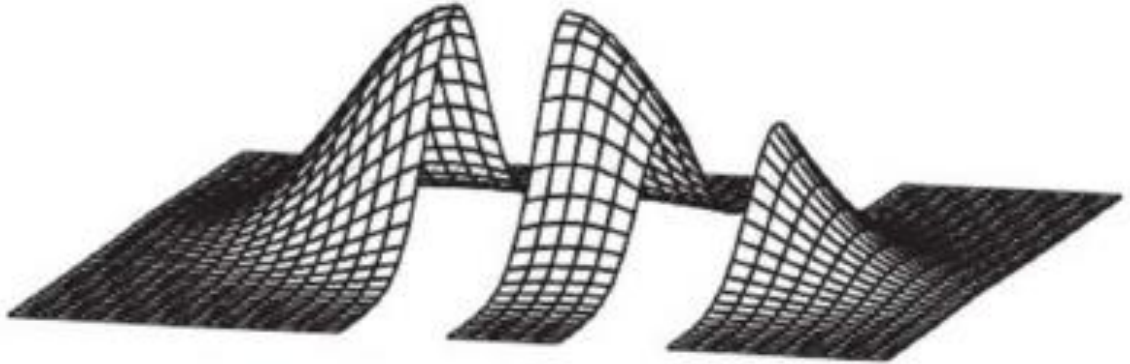
ii. The conditional distribution of X, given $Y = y$, is Gaussian and is given by

$$f_{Y|X}(y|x) = \frac{1}{\sqrt{2\pi\sigma_X^2(1-\rho_{XY}^2)}} e^{-\frac{\left[x-\mu_X-\rho_{XY}\frac{\sigma_X}{\sigma_Y}(y-\mu_Y)\right]^2}{2\sigma_X^2\left(1-\rho_{XY}^2\right)}}.$$

with mean $\mu_{X|Y} = \mu_X + \rho_{XY}\frac{\sigma_X}{\sigma_Y}(y - \mu_Y)$ and variance $\sigma_{X|Y}^2 = \sigma_X^2(1-\rho_{XY}^2)$. Similarly, the conditional distribution of Y, given $X = x$, is also Gaussian and is given by

$$f_{Y|X}(y|x) = \frac{1}{\sqrt{2\pi\sigma_Y^2(1-\rho_{XY}^2)}} e^{-\frac{\left[y-\mu_Y-\rho_{XY}\frac{\sigma_Y}{\sigma_X}(x-\mu_X)\right]^2}{2\sigma_Y^2\left(1-\rho_{XY}^2\right)}},$$

with mean $\mu_{Y|X} = \mu_Y + \rho_{XY}\frac{\sigma_Y}{\sigma_X}(x - \mu_X)$ and variance $\sigma_{Y|X}^2 = \sigma_Y^2(1-\rho_{XY}^2)$.



iii. The conditional expectation of X, given $Y = y$, $E[X|Y = y]$, is a linear function of y, given by,

$$E[X|Y = y] = a + by,$$

for some $a, b \in R$ and is given by

$$E[x|Y = y] = \mu_{X|Y} = \mu_X + \rho_{XY}\frac{\sigma_X}{\sigma_Y}(y - \mu_Y).$$

iv. The conditional variance of X, given $Y = y$, is a constant and is independent of the value of x, given by

$$\sigma^2_{X|Y} = \sigma^2_X(1 - \rho^2_{XY}).$$

We know that if X and Y are independent random variables, their correlation coefficient is 0. We also know that the converse may not be not true. However, if X and Y have a bivariate normal distribution, the converse is true as well. That is, $\rho_{XY} = 0$ implies that X and Y are independent, because, when $\rho_{XY} = 0$, $f_{XY}(x, y) = f_x(x) f_Y(y)$,

Example 8.1. At a certain university, the joint probability density function of X and Y, the grade point averages of a student in his/her freshman and senior years, respectively, is jointly Gaussian. From the grades of he past years, it is known that $\mu_X = 3$, $\mu_Y = 2.5$, $\sigma_X = 0.5$, $\sigma_Y = 0.4$ and $\rho_{XY} = 0.4$. Find the probability that a student with grade point average 3.5 in his/her freshman year will earn a grade point average of at least 3.2 in his/her senior year.

Solution. The conditional probability density function of Y, given that $X = 3.5$, is normal with mean

$$\begin{aligned}
\mu_{Y|X} &= \mu_Y + \rho_{XY}\frac{\sigma_Y}{\sigma_X}(x - \mu_X) \\
&= 2.5 + 0.4\frac{0.4}{0.5}(3.5 - 3.0) \\
&= 2.66.
\end{aligned}$$

and standard deviation

$$\begin{aligned}\sigma_{Y|X} &= \sigma_Y \sqrt{(1-\rho_{XY}^2)} \\ &= 0.4\sqrt{1-(0.4)^2} \\ &= 0.37.\end{aligned}$$

Therefore, the desired probability is calculated as follows:

$$\begin{aligned}P[Y \geq 3.2|X=3.5] &= P\Big[\frac{Y-2.66}{0.37} \geq \frac{3.2-2.66}{0.37}\Big|X=3.5\Big] \\ &= P\big[Z \geq 1.46|X=3.5\big] \\ &= 1-\phi(1.46) \\ &\approx 1-0.9279 = 0.0721.\end{aligned}$$

Bivariate Gaussian Random Vectors

Let us first assume that the two dimensional random vector is

$$\mathbf{X} = \begin{bmatrix} X_1 & X_2 \end{bmatrix}^T$$

with expectation vector

$$\boldsymbol{\mu}_{\mathbf{X}} = \begin{bmatrix} \mu_1 & \mu_2 \end{bmatrix}^T$$

and the covariance matrix

$$\boldsymbol{C}_{\mathbf{X}} = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}$$

Note that the joint Gaussian PDF of X and Y is given by

$$f_{XY}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho_{XY}^2}} e^{-\frac{\left(\frac{x-\mu_X}{\sigma_X}\right)^2 - 2\rho_{XY}\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2}{2\left(1-\rho_{XY}^2\right)}}$$

The determinant of the covariance matrix is

$$
\begin{aligned}
|\boldsymbol{C_X}| &= \sigma_1^2\sigma_2^2 - (\rho\sigma_1\sigma_2)^2 \\
&= \sigma_1^2\sigma_2^2(1-\rho^2).
\end{aligned}
$$

The inverse of the covariance matrix is

$$
\begin{aligned}
\boldsymbol{C_X}^{-1} &= \frac{\begin{bmatrix} \sigma_2^2 & -\rho\sigma_1\sigma_2 \\ -\rho\sigma_1\sigma_2 & \sigma_1^2 \end{bmatrix}}{\sigma_1^2\sigma_2^2(1-\rho^2)} \\
&= \frac{\begin{bmatrix} \sigma_1^{-2} & -\rho\sigma_1^{-1}\sigma_2^{-1} \\ -\rho\sigma_1^{-1}\sigma_2^{-1} & \sigma_2^{-2} \end{bmatrix}}{(1-\rho^2)}
\end{aligned}
$$

$$
\begin{aligned}
(\boldsymbol{x}-\boldsymbol{\mu_X})^T \boldsymbol{C_X}^{-1}(\boldsymbol{x}-\boldsymbol{\mu_X}) &= \begin{bmatrix} x_1-\mu_1 & x_2-\mu_2 \end{bmatrix} \frac{\begin{bmatrix} \sigma_1^{-2} & -\rho\sigma_1^{-1}\sigma_2^{-1} \\ -\rho\sigma_1^{-1}\sigma_2^{-1} & \sigma_2^{-2} \end{bmatrix}}{(1-\rho^2)} \begin{bmatrix} x_1-\mu_1 \\ x_2-\mu_2 \end{bmatrix} \\
&= \frac{\begin{bmatrix} \frac{x_1-\mu_1}{\sigma_1^2} - \rho\frac{x_2-\mu_2}{\sigma_2}\sigma_1^{-1} & -\rho\frac{x_1-\mu_1}{\sigma_1}\sigma_2^{-1} + \frac{x_2-\mu_2}{\sigma_2^2} \end{bmatrix}}{(1-\rho^2)} \begin{bmatrix} x_1-\mu_1 \\ x_2-\mu_2 \end{bmatrix} \\
&= \frac{\left(\frac{x_1-\mu_1}{\sigma_1}\right)^2 - \rho\frac{x_1-\mu_1}{\sigma_1}\frac{x_2-\mu_2}{\sigma_2} - \rho\frac{x_1-\mu_1}{\sigma_1}\frac{x_2-\mu_2}{\sigma_2} + \left(\frac{x_2-\mu_2}{\sigma_2}\right)^2}{(1-\rho^2)} \\
&= \frac{\left(\frac{x_1-\mu_1}{\sigma_1}\right)^2 - 2\rho\frac{x_1-\mu_1}{\sigma_1}\frac{x_2-\mu_2}{\sigma_2} + \left(\frac{x_2-\mu_2}{\sigma_2}\right)^2}{(1-\rho^2)} \qquad (8.12)
\end{aligned}
$$

Now the joint Gaussian PDF can be given by

$$f_{\boldsymbol{X}}(\boldsymbol{x}) = \frac{1}{\sqrt{(2\pi)^2 |\boldsymbol{C}_{\boldsymbol{X}}|}} e^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_{\boldsymbol{X}})^T \boldsymbol{C}_{\boldsymbol{X}}{}^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_{\boldsymbol{X}})}$$

The multivariate joint PDF of random vector $\boldsymbol{X}$ is Gaussian, if its PDF is given by

$$f_{\boldsymbol{X}}(\boldsymbol{x}) = \frac{1}{(2\pi)^{N/2}|\boldsymbol{C}_{\boldsymbol{X}}|^{1/2}} e^{-(\boldsymbol{x}-\boldsymbol{\mu}_{\boldsymbol{X}})^T \boldsymbol{C}_{\boldsymbol{X}}^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_{\boldsymbol{X}})} \tag{8.13}$$

where

$\boldsymbol{C}_{\boldsymbol{X}}$ is the covariance matrix of $\boldsymbol{X}$

$\boldsymbol{\mu}_{\boldsymbol{X}}$ is the expectation vector of $\boldsymbol{X}$, and

$|\boldsymbol{C}_{\boldsymbol{X}}|$ is the determinant of $\boldsymbol{C}_{\boldsymbol{X}}$.

Theorem 8.1. *If the component random variables of a random vector $\boldsymbol{X}$ are independent, the covariance $\boldsymbol{C}_{\boldsymbol{X}}$ is a diagonal matrix.*

Proof. If the component random variables of $\boldsymbol{X}$ are independent, then X_i and X_j are independent if $i \neq j$. Further, if the random variables X_i and X_j are independent then

$$\mathrm{Cov}[X_i, X_j] = 0.$$

$$\begin{aligned}
\boldsymbol{C}_{\boldsymbol{X}} &= \begin{bmatrix} \mathrm{Var}[X_1] & \mathrm{Cov}[X_1, X_2] & \cdots & \mathrm{Cov}[X_1, X_N] \\ \mathrm{Cov}[X_2, X_1] & \mathrm{Var}[X_2] & \cdots & \mathrm{Cov}[X_2, X_N] \\ \cdots & \cdots & \cdots & \cdots \\ \mathrm{Cov}[X_N, X_1] & \mathrm{Cov}[X_N, X_2] & \cdots & \mathrm{Var}[X_N] \end{bmatrix} \\
&= \begin{bmatrix} \sigma^2_{X_1} & 0 & \cdots & 0 \\ 0 & \sigma^2_{X_2} & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & \sigma^2_{X_N} \end{bmatrix}
\end{aligned}$$

Definition 8.2. *Mahalanobis Distance.* The quadratic form in the exponent of the PDF of the Gaussian random vector is

$$\Delta^2 = (\boldsymbol{x} - \boldsymbol{\mu_X})^T \boldsymbol{C}_{\boldsymbol{X}}^{-1} (\boldsymbol{x} - \boldsymbol{\mu_X}).$$

The quantity Δ is called the Mahalanobis distance.